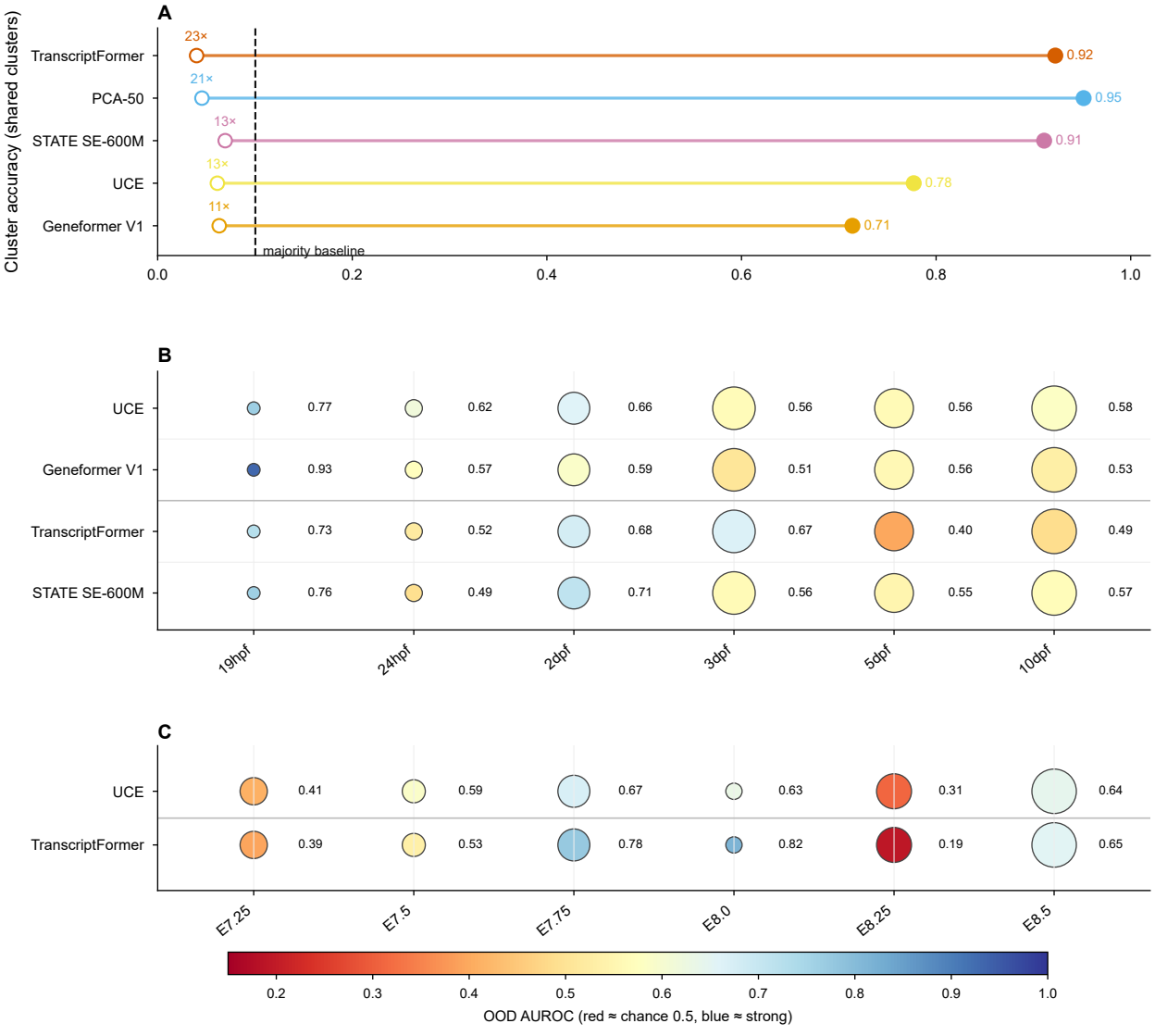
A
Cluster accuracy (shared clusters)
TranscriptFormer
PCA-50
STATE SE-600M
UCE
Geneformer V1
23×
21×
13×
13×
11×
0.92
0.95
0.91
0.78
0.71
majority baseline
0.0
0.2
0.4
0.6
0.8
1.0
B
UCE
Geneformer V1
TranscriptFormer
STATE SE-600M
0.77
0.62
0.66
0.56
0.56
0.58
0.93
0.57
0.59
0.51
0.56
0.53
0.73
0.52
0.68
0.67
0.40
0.49
0.76
0.49
0.71
0.56
0.55
0.57
19hpf
24hpf
2dpf
3dpf
5dpf
10dpf
C
UCE
TranscriptFormer
0.41
0.59
0.67
0.63
0.31
0.64
0.39
0.53
0.78
0.82
0.19
0.65
E7.25
E7.5
E7.75
E8.0
E8.25
E8.5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
OOD AUROC (red ≈ chance 0.5, blue ≈ strong)